COLLEGE OF INFORMATION TECHNOLOGY AND ENGINEERING
School Year 2025-2026

I. COURSE CODE : CpE 304

II. COURSE TITLE : Fundamentals of Mixed Signals and Sensors

III. COURSE DESCRIPTION :

This course covers operational amplifiers, signal converters, power switching devices and the construction and operation of sensors and transducers for converting physical parameters into electrical signals and vice-versa. The course focuses on the application of these devices in developing signal conversion circuits that allows measurement, processing and control of physical parameters by digital processing systems such as a finite state machine or a digital computer. Topics on actuators are also included.

Anchored in the Marian and Augustinian values of truth (veritas), unity (unitas), and charity (caritas), the course encourages students to design technology that promotes social responsibility, collaboration, and compassion. Through hands-on projects, learners will not only develop technical proficiency but also embody ethical stewardship—using their knowledge to create solutions that serve the needs of others and contribute to the common good.

IV. PRE-REQUISITE : FUNEC, FUNELC

V. CREDIT UNITS : 3 units lecture

VI. COURSE LEARNING OUTCOMES:

Vision	Mission
Moved by the Spirit of the Risen Christ, La Consolacion University Philippines, envisions itself to become a catalyst for the wholistic transformational development of Catholic Augustinian Marian graduates rooted in Gospel values.	To achieve the above vision, LCUP commits itself to generate and make available to its Augustinian Marian students, programs and services that will make them: ● *exhibit academic and technical competence in their respective areas of discipline in the local and international environment;* ● *demonstrate their knowledge, critical thinking and creativity by exploring the various avenues of learning and searching for truth;*

	● *live the Gospel values in the varied aspects of their personal and professional lives;* ● *manifest their willingness to serve specially the marginalized sector of our society;* ● *demonstrate productivity in various areas of learning as responsible citizens of the society; and* ● *develop individuals who are resilient, inclusive and responsive to be socio-economic realities of the Philippines and the complexities of the global environment.*

EXPECTED GRADUATE ATTRIBUTES	PROGRAM OUTCOMES	COURSE LEARNING OUTCOMES
EGA 1: Critical and Creative thinker EGA 2: Competent Catholic Augustinian-Marian Professional EGA 3: Socially Responsive Steward EGA 4: Transformative Lifelong learner	PO1. Ability to apply knowledge of mathematics and science to solve complex engineering problems; PO2. Ability to design and conduct experiments, as well as to analyze and interpret data; PO3. Ability to design a system, component, or process to meet desired needs within realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability and sustainability, in accordance with standards. PO4. Ability to function on multidisciplinary teams PO5. Ability to identify; formulate, and solve complex engineering problems; PO6. Understanding of professional and ethical responsibility. PO7. Ability to communicate effectively	CLO1: Identify the design process of electronic circuits and systems and conduct experiments involving sensors and mixed-signal systems, analyze data, and interpret results to improve performance and reliability. **[EGA1, PO1, PO2, PO3, PO5]** CLO2: Develop analytical and design skills for both analog and digital electronic circuit and implement sensor-based systems that integrate hardware and software components while considering economic, environmental, and ethical constraints. **[EGA2, PO6. PO7, PO12]** CLO3: Analyze and apply the behavior of passive components in AC circuits and their instrumentation applications in multidisciplinary teams to design, test, and optimize mixed-signal and sensor applications for engineering solutions **[EGA3, PO4, PO9]** CLO4: Exhibit lifelong learning, precision, and ethical responsibility in performing

	PO8. Broad education necessary to understand the impact of engineering solutions in a global, economic, environmental, and societal context. PO9. Recognition of the need for, and an ability to engage in life-long learning; PO10. Knowledge of contemporary issues; PO11. Ability to use techniques, skills and modern engineering tools necessary for engineering practice; and PO12. Knowledge and understanding of engineering and management principles as a member and leader in a team, to manage projects and in PO13: Ability to design, develop, implement and improve integrated system that include people materials, implementation, equipment and energy.	measurements and calibration tasks. Engage in continuous learning by exploring emerging technologies and trends in sensors, instrumentation, and mixed-signal design for future innovation **[EGA 4, PO11, PO13]**

VII. LEARNING PLAN:

SCHEDULE	INTENDED LEARNING OUTCOMES	LEARNING CONTENTS	SUGGESTED LEARNING ACTIVITIES	SUGGESTED ASSESSMENT TASKS	EXPECTED VALUE TO BE INTEGRATED
Week 1		**I.ORIENTATION** • Introduction to the Course Contents • Course Requirements and Activities • Course Objectives • Grading System and Policies on Absences,	• Setting and sharing of expectations from the course • Discussion of the Vision, Mission and Goals of LCUP and of CITE • Presentation of Course Objectives and Contents, Course Requirements and Activities	• Statement of expectations • Review of Course Contents, Objectives, Activities and Assessment Tools • Review of Policies on Absences, Tardiness, Examinations and Grading System	• Understanding • Open Mindedness • Respect • Obedience

		Tardiness and Examinations • Introduction to Gsuite, Google Classroom, Google Slides, Google Docs and Google Sheets	• Review of Policies on Absences, Tardiness, Examinations and Grading System		
Week 2-3	ILO1: To develop skills in analysis and design of both analog and digital circuits **[CLO1]** ILO2: Understand the design process of electronic circuits and systems. **[CLO1**]	**II. Introduction to Electronic System** • History of Electronics • Electronic Signals and Notation • Classifications of Electronic Systems • Specifications of Electronic Systems • Design of Electronic Systems • Design of Electronic Circuits • Electronic Devices • Emerging Electronics **Differential Amplifiers** • Introduction to Differential Amplifiers • Block Diagram • Differential Gain • Common Mode Gain	• Lecture-Discussion using PowerPoint	• Quiz on the topics discussed	• Cooperation • Understanding • Critical Thinking • Active participation and involvement

		• CMRR • Configuration of differential amplifiers			
Week 4-5	ILO1: Show one's knowledge on the fundamental elements of amplifiers.**[CLO1, CLO2]** ILO2: Refresh and expand the understanding of basic electrical components and the basic terms used in electricity as required for instrumentation. .**[CLO1, CLO2]**	**III. INTRODUCTION TO OPERATIONAL AMPLIFIERS AND APPLICATIONS** • Characteristics of Ideal Op-Amps • Noninverting Amplifiers • Inverting Amplifiers • Instrumentation Amplifiers • Integrators • Differentiators • Addition–Subtraction Amplifiers • Large-Signal Operations of Op-Amps • Input Offset Voltage • Input Offset Current • Op-Amp Circuit Design **Linear and Non Linear Applications of Op-Amp** • Linear Op-amp • Non-linear Op-amp	• Lecture-Discussion using PowerPoint	• Quiz on the topics discussed	• Cooperation • Understanding • Critical Thinking • Active participation and involvement

Week 6		**Preliminary Examination**	• Examination on the topics discussed during the Mid Term	• Examination	• Honesty in answering the test questionnaire • Diligence • Understanding
Week 7-8	ILO1: To develop skills in analysis and design of both analog and digital circuits ILO6: Understand active devices, how they are used in instrument applications, and the difference between analog and digital circuits. ILO10: Demonstrate the Filipino-Catholic Augustinian Spirituality through value webbing.	**IV. Analog to Digital Conversion** **Digital to Analog Conversion** • Sampling • Quantization • encoding	• Lecture-Discussion using PowerPoint	• Quiz on the topics discussed	• Cooperation • Understanding • Critical Thinking • Active participation and involvement
Week 9-11	ILO4: Refresh and expand the understanding of basic electrical components and the basic terms used in electricity as required for instrumentation. ILO7: Understand the units used in measurements and familiarize with the most common	**V. Mixed Signals** **Basic Measurement Theory** • IC Test • Linearity	• Lecture-Discussion using PowerPoint	• Quiz on the topics discussed	• Cooperation • Understanding • Critical Thinking • Active participation and involvement

	methods of using the various measurement . ILO10: Demonstrate the Filipino-Catholic Augustinian Spirituality through value webbing.				
Week 12		**Midterm Examination**	• Examination on the topics discussed during the Mid Term	• Honesty in answering the test questionnaire • Diligence • Understanding	• Examination
Week 13-14	ILO8: Understand the different Sensors and Transducers. ILO10: Demonstrate the Filipino-Catholic Augustinian Spirituality through value webbing	**VI. Sensors and Transducers** ● Sensitivity & Hysteresis ● Offset Error & Non-linearity	• Lecture-Discussion using PowerPoint	• Quiz on the topics discussed	• Cooperation • Understanding • Critical Thinking • Active participation and involvement
Week 15-16	ILO8: Understand the different Sensors and Transducers.	**VII. Other Types of Sensors and Transducers** ● active sensors ● passive sensors	• Lecture-Discussion using PowerPoint	• Quiz on the topics discussed	• Cooperation • Understanding • Critical Thinking • Active participation and involvement
Week 17	ILO9: Learn the basic components and control of devices and prototypes. ILO10: Demonstrate the Filipino-Catholic Augustinian Spirituality through value webbing	**VIII. Basic Control Devices** ● What are control devices ● different types of control devices	• Lecture-Discussion using PowerPoint	• Quiz on the topics discussed	• Cooperation • Understanding • Critical Thinking • Active participation and involvement

Week 18		**Final Examination**	• Examination on the topics discussed during the Final Term	• Final Examination	• Honesty in answering the test questionnaire • Diligence

VIII. COURSE REQUIREMENTS :

1. Class Attendance
2. Recitations
3. Quizzes
4. Examinations
5. Compilation of Laboratory Activity Reports
6. Project

IX. REFERENCES :

A. INTERNET REFERENCES:

Erhan, L., Ndubuaku, M., Di Mauro, M., Song, W., Chen, M., Fortino, G., Bagdasar, O., & Liotta, A. (2020). *Smart Anomaly Detection in Sensor Systems: A Multi-Perspective Review*. arXiv. https://arxiv.org/abs/2010.14946 arXiv

Wei, Z., Qu, H., Wang, Y., Yuan, X., Wu, H., Du, Y., Han, K., & Zhang, N. (2023). *Integrated Sensing and Communication Signals Toward 5G-A and 6G: A Survey*. arXiv. https://arxiv.org/abs/2301.03857 arXiv

Cherubini, A., & Navarro-Alarcon, D. (2020). *Sensor-Based Control for Collaborative Robots: Fundamentals, Challenges and Opportunities*. arXiv. https://arxiv.org/abs/2007.02067 arXiv

Perera, C., Zaslavsky, A., Christen, P., & Georgakopoulos, D. (2013). *Context Aware Computing for the Internet of Things: A Survey*. arXiv. https://arxiv.org/abs/1305.0982 arXiv

X. GRADING SYSTEM :

1. Class Standing 60%
2. Term Examination 40%
3. Term Grade = Class Standing (60%) + Term Examination (40%)
4.
5. Final Rating = <u>Preliminary Term Grade + Mid Term Grade + Final Term Grade</u>

Prepared by: **ENGR. GLAZEN CAREY PEREZ** Faculty/Date	Checked and Reviewed by: **EDNA F. DAYAO, DIT** Dean/Date **ROSALYN S. GALVEZ, PhD., LPT** Asst. Vice President for Academic Affairs-Date References Certified by: **MARIA MERCEDEZ M. HAZ, PhD., RL** LIC Director/Date	Approved by: **JAY MARK B. SANTOS, PhD, LPT, CSSWB** Vice President for Academic Affairs-Date

Revision Date:	
Modifications: __Learning Outcomes __Course Contents __Suggested Learning Activities __Suggested Assessment __Expected Values to be Integrated __Course Requirements __References __Grading System	